

Customer Churn Prediction & LTV Engine

Week 4 • Day 1

Dashboard Development — Superset/Metabase Setup

Install and configure Apache Superset or Metabase and connect to PostgreSQL

■ Day Objectives

- ◆ Choose between Apache Superset and Metabase for your dashboard
- ◆ Install and start the chosen tool
- ◆ Connect the tool to your churn_db PostgreSQL database
- ◆ Navigate the dashboard interface
- ◆ Create your first data table and chart

■ 1. Superset vs Metabase

Apache Superset is an open-source enterprise BI tool with many chart types and SQL Lab. More powerful but more complex to configure.

Metabase is simpler, has a cleaner UI, and requires no SQL knowledge for basic dashboards. Recommended for beginners.

■■ **Note:** We recommend Metabase for this project due to its simpler setup. All instructions below cover Metabase.

■ 2. Metabase Setup via Docker

■ Install and start Metabase with Docker

```
# First, make sure Docker is installed (covered in Day 25)
# For now, use the JAR file approach (no Docker needed):

# Download Metabase JAR:
# Go to https://www.metabase.com/start/oss/
# Download metabase.jar

# Run Metabase (requires Java 11+):
# Install Java: https://adoptium.net/
java -jar metabase.jar

# Metabase starts at: http://localhost:3000
# Setup wizard appears on first run
```

■ Metabase initial setup

```
# 1. Open http://localhost:3000
# 2. Click 'Let's get started'
# 3. Create admin account with your email/password
# 4. On 'Add your data' screen:
#   - Database type: PostgreSQL
#   - Name: Churn Analytics DB
#   - Host: localhost
#   - Port: 5432
#   - Database name: churn_db
#   - Username: postgres
#   - Password: postgres123
#   - Click 'Test Connection' → should show green checkmark
#   - Click 'Save'
# 5. Click 'Take me to Metabase'
```

■ 3. Commit & Checklist

- Metabase running at http://localhost:3000
- PostgreSQL connection test passes (green checkmark)
- Can browse customers_ltv table in Metabase

■ Week4-Day1 Commit

```
git add .
git commit -m "Week4-Day1: Metabase installed, PostgreSQL connection configured"
git push origin main
```

■ Estimated Time: 4 – 5 hours